

FAREWELL SPIT, New Zealand

WMO: 935270

Lat: 40.545S

Lon: 173.001E

Elev: 5

StdP: 101.27

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	5.4	-0.8	3.5	8.4	0.9	4.0	7.6	16.5	11.4	14.9	11.4	2.8	260	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.2	23.3	18.6	22.4	18.0	21.5	17.5	19.7	21.9	19.1	21.1	18.5	20.5	6.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	20.6	18.3	13.2	20.2	17.6	12.6	19.7	56.4	22.0	54.3	21.2	52.3	20.5	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	13.1	11.9	3.0	25.4	1.6	1.1	1.9	26.3	0.9	26.9	0.0	27.6	-1.1	28.4
			1.5	20.7	1.1	0.8	0.8	21.3	0.2	21.8	-0.4	22.2	-1.2	22.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	17.8	18.2	17.1	15.0	13.1	10.9	10.1	10.6	11.9	13.1	14.5	16.4	(6)
(7)		DBStd	3.22	1.77	1.59	1.67	1.66	1.87	1.95	1.70	1.60	1.50	1.52	1.65	1.72	(7)
(8)		HDD10.0	49	0	0	0	0	2	13	19	12	3	1	0	0	(8)
(9)		HDD18.3	1606	31	19	45	99	161	222	255	239	193	162	114	63	(9)
(10)		CDD10.0	1526	242	230	220	151	98	40	23	32	59	97	136	198	(10)
(11)		CDD18.3	41	14	16	7	0	0	0	0	0	0	0	1	3	(11)
(12)		CDH23.3	31	16	10	2	0	0	0	0	0	0	0	0	3	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.0	5.5	5.4	5.1	5.5	5.5	5.5	6.2	6.7	6.9	6.2	(14)	
(15)	Precipitation	PrecAvg	1528	109	124	103	137	125	143	146	159	129	140	120	108	(15)
(16)		PrecMax	2302	201	368	169	276	263	245	300	358	273	289	248	290	(16)
(17)		PrecMin	1067	33	43	44	21	7	51	44	60	38	52	30	40	(17)
(18)		PrecStd	295	48	81	37	75	58	54	64	59	56	62	66	60	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	24.6	23.2	20.5	18.6	16.2	15.1	15.6	17.3	19.2	21.7	23.5	(19)
(20)			MCWB	19.6	19.4	18.6	16.8	15.8	14.2	12.9	12.7	13.4	14.7	16.3	18.1	(20)
(21)		2%	DB	23.3	23.3	22.0	19.4	17.6	15.2	14.4	14.8	16.1	17.8	19.8	21.7	(21)
(22)			MCWB	18.7	18.7	17.8	16.0	15.2	13.5	12.4	12.2	13.0	14.1	15.3	17.2	(22)
(23)		5%	DB	22.2	22.4	21.1	18.6	16.8	14.7	13.7	14.1	15.4	16.9	18.7	20.6	(23)
(24)			MCWB	17.9	18.2	17.3	15.6	14.7	13.0	11.9	11.8	12.6	13.5	14.7	16.6	(24)
(25)		10%	DB	21.2	21.6	20.3	17.9	16.1	14.1	13.1	13.5	14.7	16.1	17.8	19.6	(25)
(26)			MCWB	17.4	17.6	16.7	15.2	14.1	12.4	11.2	11.4	12.2	13.0	14.2	16.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	20.7	19.8	18.3	17.1	15.1	14.1	14.0	14.7	15.9	17.4	19.2	(27)
(28)			MCDWB	23.8	23.0	21.9	19.2	17.7	15.5	14.4	14.7	15.7	17.4	19.9	21.9	(28)
(29)		2%	WB	19.7	19.8	19.0	17.3	16.2	14.2	13.1	13.1	14.1	15.1	16.4	18.2	(29)
(30)			MCDWB	22.1	21.8	20.7	18.4	16.9	14.9	13.8	14.0	15.1	16.5	18.6	20.3	(30)
(31)		5%	WB	19.0	19.2	18.2	16.6	15.4	13.6	12.2	12.5	13.6	14.4	15.7	17.6	(31)
(32)			MCDWB	20.9	21.2	20.0	17.7	16.4	14.4	13.2	13.6	14.6	15.9	17.6	19.5	(32)
(33)		10%	WB	18.3	18.5	17.5	16.0	14.6	12.8	11.4	11.8	13.0	13.8	15.1	17.0	(33)
(34)			MCDWB	20.4	20.7	19.4	17.2	15.8	13.8	12.6	13.1	14.2	15.3	16.7	18.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.2	6.1	5.7	5.2	5.1	5.2	5.5	5.5	5.9	6.0	5.9	(35)
(36)			MCDBR	7.3	7.1	6.8	6.2	5.6	5.0	5.4	5.8	6.1	6.7	7.0	6.9	(36)
(37)			MCWBR	4.1	4.1	3.9	4.1	4.0	4.0	4.1	4.1	4.1	4.4	4.1	4.1	(37)
(38)		5% WB	MCDBR	6.0	5.8	5.6	4.7	4.3	3.9	4.6	4.9	5.0	5.4	5.5	5.6	(38)
(39)			MCWBR	3.8	3.9	3.8	3.7	3.8	3.6	4.0	4.0	4.1	4.2	3.8	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.336	0.322	0.314	0.302	0.301	0.294	0.304	0.333	0.338	0.332	0.341	(40)	
(41)		taud	2.576	2.621	2.661	2.666	2.660	2.634	2.656	2.643	2.539	2.545	2.586	2.595	(41)	
(42)		Ebn at Noon	984	968	941	884	830	795	831	881	912	955	991	993	(42)	
(43)		Edh at Noon	104	95	83	72	62	58	61	72	92	101	102	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.89	6.06	4.74	3.16	2.16	1.64	1.88	2.70	3.88	5.18	6.42	6.68	(44)	
(45)		RadStd	0.46	0.48	0.22	0.28	0.10	0.10	0.12	0.15	0.25	0.41	0.50	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.45	+1.32	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page